

Jonathan Chong

📞 403-472-2006 | ✉ jonathanchongyyc@gmail.com | 🔗 linkedin.com/in/jonathanchongyyc | 🌐 github.com/wejona

EDUCATION

University of Calgary

Bachelor of Science in Software Engineering, Minor in Biomedical Engineering

Graduated May 2025

GPA: 3.8/4.0

SKILLS

Technical Skills: Artificial Intelligence, Full Stack Web Development, Software Architecture, Computer Architecture, Database Architecture

Programming Languages: Python, C++, C, Java, SQL, JavaScript, TypeScript, HTML, CSS

Libraries/Frameworks: React, Node.js, Flask, Express, Axios, TensorFlow, Scikit-learn, LangChain

Developer Tools: Git, GitHub, Gerrit, Vim, VSCode, Docker, AWS, Microsoft Azure, Ansible, Jira, Linux

WORK EXPERIENCE

Software Development Engineer Intern

Advanced Micro Devices Inc. (AMD)

May 2023 – August 2024

Toronto, Ontario

- Architected a **centralized database** consolidating data from critical internal tools and trained a model to **automate failure classification** with **93% accuracy** using **SQL** and **TensorFlow**.
- Deployed a **full-stack web platform** to log test failures, analyze failure rates, predict build times, and surface software defects, **eliminating dozens of hours** of manual triage using **Snowflake, SQL, React, Flask, and Ansible**.
- Engineered a **front-end UI** and **back-end search engine** that replaced a legacy workflow for over **350 engineers**, built with **SQL, Python, Flask, Axios, React, JavaScript, and MUI**.
- Leveraged **Large Language Models (LLMs)** enhanced with **Retrieval Augmented Generation (RAG)** over Confluence and Jira APIs to **automate ticket creation** using **Azure OpenAI, LangChain, and Python**.
- Established the **CI/CD pipeline** for over **10 CPU, GPU, and NPU products** used by **350+ engineers** through **Linux, Buildbot, and Python**.
- Streamlined the **development and release process** across multiple programs by hardening a shared interface tool in **Linux and Python**, **saving hundreds of engineering hours** in troubleshooting.
- Maintained and contributed to **multiple repositories** responsible for **ASIC and firmware qualification, IP design verification, and benchmarking** across **10+ GPU, CPU, and NPU programs**.

Software Engineer Intern

Garmin Ltd.

Jan 2022 – Dec 2022

Calgary, Alberta

- Launched a **Python module** for transmitting and receiving **low-level data with embedded devices**, adopted by **50+ engineers** and integrated as a core component across **6+ repositories**.
- Automated **Bluetooth Low Energy (BLE) protocol** testing between sensors and consumer products using **C++**.
- Refactored a legacy test application to support modern **Android/iOS devices** and additional mobile apps using **Selenium and pytest**, increasing automated test coverage by over **60%**.
- Applied **Test Driven Development (TDD)** to build a framework for controlling modern and legacy applications via **REST APIs and Python**.
- Built internal tooling, including a **mobile app and log-retrieval script**, that **eliminated hours** of repetitive work using **Kotlin, Java, and Bash**.

Software Developer

Neocycle (Startup)

Jan 2021 – Dec 2021

Calgary, Alberta

- Spearheaded a synthetic biology solution to recover rare earth elements from electronic waste, **reducing environmental impact by 80%** and **costs by 30%**.
- Designed a **random forest model** with **nested cross-validation** to predict kinetic rate constants for a target protein-ligand interaction with **95% accuracy**.
- Engineered novel protein sequences that **improved binding affinity by 25%** using **convolutional neural networks (CNNs)** and additional **machine learning** techniques.
- Secured over **\$25,000** in startup funding through business and research competitions.

PROJECTS

Bloombot – A Smart Garden System *powered by Symend*

Sept 2024 – April 2025

- Designed a fully automated smart-home gardening device with pH control, enabling **automatic nutrient dosing** driven by live **sensor readings**; awarded **Best Electrical & Software Engineering Project** by the University of Calgary.
- Developed **embedded firmware** for the **ESP32-S3 microcontroller** in **C++** and **ESP-IDF**, implementing **sensor acquisition**, **actuator control**, and real-time **MQTT communication** with **AWS IoT Core**.
- Shipped **cross-platform user interfaces** in **React Native** (mobile) and **React** (web) for real-time sensor monitoring, automation rule configuration, and remote actuator control.
- Implemented **Bluetooth Low Energy (BLE) provisioning** for secure transfer of **Wi-Fi credentials** from the mobile app to the device, streamlining the user onboarding experience.
- Architected a **serverless backend** on **AWS Lambda (Node.js)**, **DynamoDB**, and **AppSync (GraphQL)** for bi-directional device control and real-time data flow.
- Authored **actuator drivers** for **pumps, fans, and lighting** using **PWM** and **GPIO**, achieving sub-100 ms response latency to automation commands.

Rating Rager – Data-Driven Restaurant Ranking System

Sept 2024 – Dec 2024

- Designed a dynamic weighted ranking system for every Calgary restaurant by combining sentiment, trend, and geographic analysis with overall rating, uniqueness, and price.
- Engineered a data pipeline against the **Google Maps API** processing **16,274 requests** to gather and refine data on all **3,940** Calgary restaurants.
- Applied **VADER** sentiment analysis to customer reviews to produce accurate, timely sentiment scores.
- Used clustering algorithms (**KNN** and **DBSCAN**) to analyze geographic sentiment and identify under-served areas and high-performing restaurant clusters.
- Classified cuisine types from restaurant data via **Natural Language Inference (NLI)** with **92% accuracy**, filling gaps in API-provided data.

Macro Pad from Scratch

Dec 2023 – Jul 2024

- Designed a custom PCB in **KiCad** and manufactured it via **JLCPCB**.
- Programmed **firmware** for an **AVR microcontroller** in **C** using the **LUFA** USB library.
- Modeled and 3D-printed custom rotary knobs using **SolidWorks** and **Prusa Slicer**.

I'm Slime – Video Game

Sept 2022 – Dec 2022

- Developed a **Player-versus-Player (PvP)** web game in **React** on a flexible **microservices-based architecture**.
- Built community-driven **front-end** and **back-end** features including account creation, friends list, and private/public messaging using **Firebase**, **React**, and **Axios**.
- Implemented an in-game shop with character and profile customization using **Bootstrap**, **HTML**, and **CSS**.

StudyHub: Connecting Students to Freelance Tutors

Sept 2022 – Dec 2022

- Built a **full-stack** marketplace following **Model-View-Controller (MVC)** principles to connect students with tutors.
- Developed **REST API routes** and a comprehensive database layer using **Node.js**, **Axios**, **Express.js**, and **SQL**.
- Designed a user-friendly website using **HTML**, **CSS**, **JavaScript**, and **GitHub Pages**.

CLUBS

Vice President of Technology – Genetics and Genomics Club

Sept 2020 – Jan 2022

- Built and maintained the club website with weekly announcements using **HTML**, **CSS**, and **Bootstrap**.

Vice President – Competitive Programming Club

Sept 2020 – Jan 2022

- Created and delivered weekly lessons on data structures, algorithms, and runtime & memory efficiency.

President – Calgary Developer Students Club *powered by Google*

Sept 2020 – May 2021

- Hosted workshops on industry tools such as **Firebase** and **Google Cloud**.
- Organized expert panels with leaders across computer science and software engineering.
- Created networking opportunities through career fairs and info nights.

VOLUNTEERING

- AMD × Princess Margaret Foundation** 2023 & 2024
- Volunteered as a VR Specialist at Canada's FanExpo, helping raise over **\$20,000** for cancer research.
 - Represented AMD at the Call of Duty League Toronto Ultra Major, helping raise over **\$50,000** for cancer research.
- AMD × Canada Computers Esports Arena** 2023
- Hosted and served as primary commentator for the Fortnite tournament broadcasted on the AMD Twitch channel with a **\$5,000** prize pool.
- Calgary & Alberta Collegiate Programming Contests** 2020 & 2021
- Lead event coordinator, liaising with sponsors (Arcurve, Synopsys), platform provider Kattis, and the University of Calgary for logistics.
 - Authored contest problems in LaTeX with reference solutions in C++ (fastest) and Python (slowest), verified through Docker and Kattis' Linux toolkit.
 - Achieved **200+ attendees** and **\$8,000** and **\$10,000** prize pools across the two events.

ACHIEVEMENTS

AMD Executive Spotlight Award (\$1,500)
Society of Asian Scientists & Engineers Scholarship (\$5,000)
Schulich School of Engineering Award (\$6,000)
Jason Lang Scholarship (\$4,000)
Schulich School of Engineering Dean's List (2019-2023)

COURSES & CERTIFICATIONS

Advanced C++20 Training Course at AMD
The Complete Web Development Bootcamp on Udemy
Machine Learning A-Z: AI, Python, R on Udemy
iOS & Swift - The Complete iOS App Development Bootcamp on Udemy
The Complete Python Pro Bootcamp on Udemy